CRSS

Compliance Readiness Support System

AI-powered regulatory intelligence for medtech startups navigating EU law

Diego Barra · Gründungstipendium NRW · June 2026

The Problem

EU medical device companies face **3+ major overlapping regulations**:

Regulation	Applies to	In force since
MDR 2017/745	All medical devices	May 2021 / 2024
IVDR 2017/746	In vitro diagnostics	May 2022 / 2025
EU AI Act 2024/1689	AI-enabled devices	Aug 2026 →
GDPR 2016/679	Patient data processing	May 2018

> “ *A startup building an AI-powered diagnostic tool must simultaneously comply with **all four** — each referencing the others.* ”

The Problem I Lived

I didn't start with regulations. I started with a cancer detection model.

At a medtech startup building AI for **early breast cancer detection** using Raman Spectroscopy, I found a critical flaw in our model validation pipeline:

- Patient samples were present in **both training and test sets**
- Model accuracy was **artificially inflated** — severe overfitting disguised as performance
- I raised it internally. It wasn't resolved. I left.

Then I investigated the literature — the same error was **widespread across published Raman Spectroscopy cancer detection studies**. Medical AI reporting guidelines existed. They were being ignored.

> “ ***When these research-stage models hit EU AI Act conformity assessment: the gap between how they were validated and what the law requires will be enormous. That is the market.*** ”

Why This Is Getting Worse

- **MDR/IVDR transition** is still ongoing — thousands of legacy devices still seeking CE marks under stricter rules
- **EU AI Act** obligations for high-risk AI (medical devices = automatic high-risk) take effect **August 2026**
- **Notified Body bottleneck** — only ~22 NBs designated under MDR; wait times of 18–36 months
- **Regulatory affairs talent** is scarce and expensive

> “ *The average cost of CE marking for a Class II medical device:*
> ***€100,000 – €500,000*** *in regulatory consulting alone* ”

Getting it wrong means market exclusion.

Who Needs This

Primary customer: early-stage medtech & digital health startups

- Pre-CE-marking phase, figuring out what applies to their device
- No dedicated regulatory affairs team yet
- Founders spending weeks reading legislation to answer basic questions

Secondary: regulatory affairs professionals

- Mid-size medtech companies cross-referencing obligations across regulations
- Contract regulatory consultants serving multiple clients

Market context: 27,000+ medical device companies in the EU.
NRW alone hosts major clusters in **Düsseldorf, Cologne, and Aachen**.

The Solution

CRSS is a regulatory intelligence layer built on the actual law.

A question like:

> “ *"What does the EU AI Act require from a manufacturer of a Class IIb medical device that incorporates an AI model?"* ”

Returns a **precise, cited answer** tracing obligations across MDR Article 10, AI Act Article 16, and the relevant MDCG guidance — in seconds.

Not a chatbot. Not a search engine.
The system reasons over a structured knowledge graph of the regulations themselves.

Demo

[Screenshot: question → cited answer across MDR + AI Act]

> “ *"Article 10(1) MDR requires manufacturers to establish a quality management system. Article 16(a) EU AI Act additionally requires providers of high-risk AI systems to establish a risk management system pursuant to Article 9. MDCG 2025-6 clarifies that these obligations are cumulative and complementary..."* ”

Real article numbers. Real legal text. No hallucination.

Why It Works

Three things that make this different from ChatGPT + a PDF:

1. **The full regulations are parsed and cross-linked** — every article, paragraph, and annex reference is resolved. The system knows that MDR Annex I delegates to Annex II which is referenced in Article 52.
2. **Cross-regulation reasoning** — a single question can pull obligations simultaneously from MDR, AI Act, and GDPR because the graph tracks which articles in one regulation cite which in another.
3. **Grounded answers only** — the system is architecturally prevented from citing anything not in the retrieved legal text. No training-data contamination.

Differentiation

	CRSS	Generic LLM	Regulatory Consultant	Static Compliance Tool
Grounded in actual law	✅	❌	✅	Partial
Cross-regulation reasoning	✅	❌	✅	❌
AI Act ready	✅	❌	Depends	❌
Available 24/7	✅	✅	❌	✅
Cost per question	Cents	Cents	€200–500/hr	—

The gap between a €300/hr regulatory lawyer and a well-grounded AI answer is **closing fast** — but only if the AI is actually grounded.

Business Model

SaaS, subscription-based

Tier	Target	Price (indicative)
Startup	≤10 employees, pre-CE	€149/month
Professional	Regulatory affairs teams	€499/month
Enterprise	Consultancies, NBs	Custom

Additional angles:

- White-label for regulatory consultancies (they deliver it to their clients)
- Integration into existing QMS tools (Greenlight Guru, Qualio)
- Regulation update notifications when new MDCG guidance is published

Gross margin potential: >85% (marginal cost = API calls + compute)

Traction & Validation

What exists today:

- Working prototype covering MDR, IVDR, EU AI Act, GDPR + 9 MDCG guidance documents
- Structured knowledge graph: 7,000+ legal provisions, cross-referenced
- Real Q&A capability validated on regulatory test cases

Next steps with stipend:

- 5 pilot conversations with NRW-based medtech founders (already 2 warm introductions)
- Refine pricing and onboarding flow based on pilot feedback
- Expand to German-language regulations (MDR/IVDR German text)

This is a seed-stage project. I am not claiming a validated business — I am claiming a validated problem and a working technical solution.

Why NRW

NRW is the right place to build this:

- **Düsseldorf**: home to major medtech and pharma companies (Henkel, Evonik, many device companies)
- **Cologne/Bonn**: digital health ecosystem, StartupDock, multiple health-tech accelerators
- **Aachen**: RWTH spin-off culture, strong regulatory research groups

Personal connection: Based in NRW, with access to the local medtech network through [relevant connection/university/accelerator].

The Gründungstipendium NRW gives me 12 months to stop splitting focus between building and consulting — and go full-time on customer development and product.

The Ask

12 months to validate and launch CRSS commercially

Milestones:

- Month 3: 5 paying pilots, pricing confirmed
- Month 6: German-language expansion, first enterprise conversation
- Month 9: 20 active users, first retention data
- Month 12: Seed round or revenue-sustainable

About Me

Diego Barra — ML engineer with deep medtech domain experience

- Built AI diagnostics for **early breast cancer detection** (Raman Spectroscopy) at a medtech startup
- Identified systemic validation gaps in medical AI research; investigated medical AI reporting standards and the EU regulatory framework for AI-enabled devices
- Designed and built CRSS end-to-end: knowledge graph (Neo4j), embeddings, LLM reasoning layer, cross-regulation chain retrieval

Why me specifically:

The combination of hands-on AI development in a regulated clinical context, regulatory self-study depth, and the software engineering capability to build the full system is rare.
I am not pitching an idea — the system exists.

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